7: STATISTICAL INFERENCE – POINT ESTIMATORS

Lecturer: Flemming Christensen

INSTITUT FOR ELEKTRO- OG COMPUTERTEKNOLOGI, AARHUS UNIVERSITET

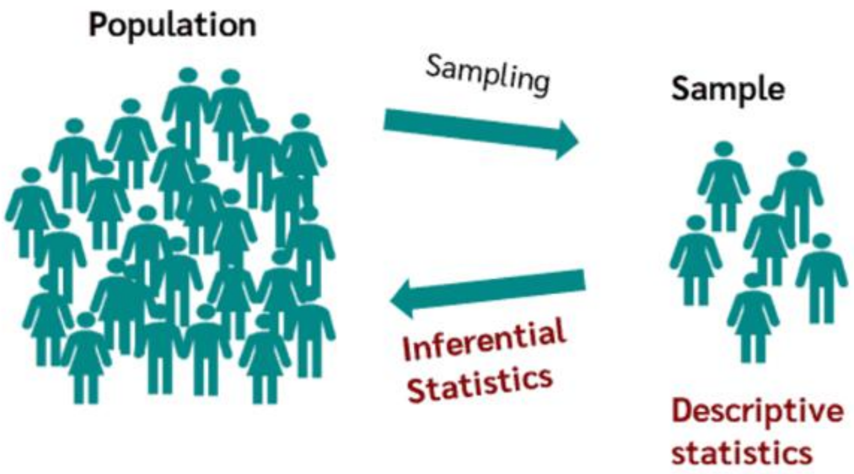
Email: flech@ece.au.dk

Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed., https://www.probabilitycourse.com

Course: Stokastisk modellering og behandling (SWSTS-01)

Descriptive Statistics vs Inferential Statistics



Feature	Descriptive Statistics	Inferential Statistics
Purpose	Summarize data	Make predictions
Data Scope	Focuses only on the given data	Generalizes to a larger population
Use of Samples	May describe a sample or population	Always uses a sample to infer about the population
Techniques	Measures of central tendency, dispersion, visualization	Hypothesis testing, confidence intervals, regression analysis
Example	Mean age of students in a class	Predicting average age of all students in the country

Independent and Identically Distributed Random Variables (i.i.d.)

In many statistical analyses, we often work with random variables that are:

- **Independent**, and
- **Identically distributed**, meaning they share the same cumulative distribution function (CDF).

This combination—independence and identical distribution—greatly simplifies analysis and is common in practice.

Definition

A collection of random variables $X_1, X_2, \ldots, X_n$ is said to be **independent and identically distributed (i.i.d.)** if both of the following hold:

1. **Independence**:

   For any real numbers $x_1, x_2, \ldots, x_n$,

$$P(X_1 \leq x_1, X_2 \leq x_2, \ldots, X_n \leq x_n) = \prod_{i=1}^{n} P(X_i \leq x_i).$$

2. **Identical Distribution**:

   For all real numbers x,

$$P(X_1 \leq x) = P(X_2 \leq x) = \cdots = P(X_n \leq x).$$

Central Limit Theorem (CLT)

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with

- $E[X_i] = \mu$
- $\mathrm{Var}(X_i) = \sigma^2$

Define the **sum**:

$$Y = X_1 + X_2 + \cdots + X_n.$$

Mean and variance of the sum

Expectation

$$E[Y] = E[X_1 + \cdots + X_n] = E[X_1] + \cdots + E[X_n] = n\mu.$$

Variance (using independence)

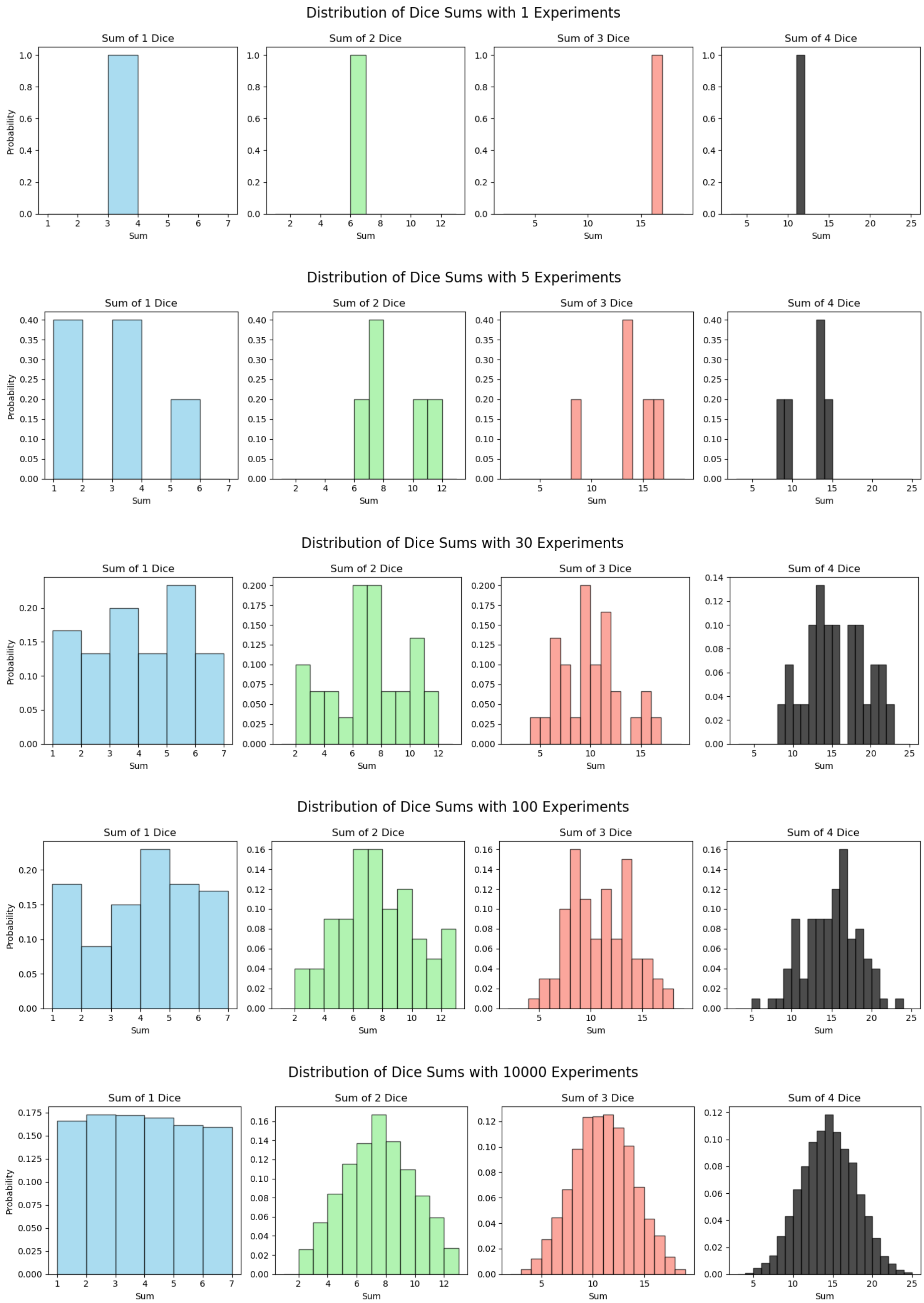
$$\mathrm{Var}(Y) = \mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n) = n\sigma^2.$$

Theorem: Central Limit Theorem

For large n, the distribution of Y is approximately normal:

$$Y \approx N(n\mu,\ n\sigma^2).$$

CLT and LLN



LLN

CLT

Central Limit Theorem (CLT)

Suppose we want to transform

$$Y \approx N(n\mu,\ n\sigma^2)$$

into a standard normal distribution

$$Z_n \approx N(0,1)$$

Apply inverse linear transformation:

$$Y_n = \sigma\sqrt{n} \cdot Z_n + n\mu \ \Rightarrow \boldsymbol{Z_n} = \frac{\boldsymbol{Y_n - n\mu}}{\boldsymbol{\sigma\sqrt{n}}}$$

So

$$Z_n \xrightarrow{d} N(0,1) \quad \text{as } n \to \infty,$$

equivalently,

$$P(Z_n \leq z) \to \Phi(z), \qquad n \to \infty$$

where Φ is the CDF of $N(0,1)$.

Standardized CLT allows probability approximation

From

$$P(Z_n \leq z) \to \Phi(z)$$

And

$$P(a \leq Z_n \leq b) = P(Z_n \leq b) - P(Z_n \leq a)$$

We get

$$P(a \leq Z_n \leq b) \approx \Phi(b) - \Phi(a)$$

for large n.

Central Limit Theorem (CLT)

How to Apply the Central Limit Theorem (CLT)

1. **Define the variable of interest**

   Let $Y = X_1 + X_2 + \cdots + X_n$, where $X_1, X_2, \ldots, X_n$ are i.i.d. random variables.

2. **Compute the mean and variance**

   If $E[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$, then

   $$E[Y] = n\mu, \quad \mathrm{Var}(Y) = n\sigma^2.$$

3. **Standardize and apply the CLT**

   By the Central Limit Theorem,

   $$Z = \frac{Y - n\mu}{\sqrt{n}\sigma}$$

   is approximately standard normal for large n.

4. **Transform the probability statement**

   To find $P(y_1 \le Y \le y_2)$, subtract $n\mu$ and divide by $\sqrt{n}\sigma$ throughout:

   $$P(y_1 \le Y \le y_2) = P\left(\frac{y_1 - n\mu}{\sqrt{n}\sigma} \le \frac{Y - n\mu}{\sqrt{n}\sigma} \le \frac{y_2 - n\mu}{\sqrt{n}\sigma}\right)$$

5. **Approximate using the standard normal CDF**

   Since $\frac{Y - n\mu}{\sqrt{n}\sigma} \approx Z \sim N(0,1)$,

   $$P(y_1 \le Y \le y_2) \approx \Phi\left(\frac{y_2 - n\mu}{\sqrt{n}\sigma}\right) - \Phi\left(\frac{y_1 - n\mu}{\sqrt{n}\sigma}\right)$$

Central Limit Theorem (CLT)

Example

In a communication system each data packet consists of 1000 bits. Due to the noise, each bit may be received in error with probability 0.1. It is assumed bit errors occur independently. Find the probability that there are more than 120 errors in a certain data packet and less than 130 errors.

Let X_i indicate an error on bit i in a 1000-bit packet:

$$X_i = \begin{cases} 1, & \text{bit } i \text{ is in error} \\ 0, & \text{otherwise} \end{cases} \qquad X_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p = 0.1).$$

Let $Y = \sum_{i=1}^{1000} X_i$ be the number of bit errors. Then

$$E[Y] = np = 100, \qquad \text{Var}(Y) = np(1-p) = 90, \qquad \sigma = \sqrt{90}.$$

By the CLT,

$$Z = \frac{Y - 100}{\sqrt{90}} \approx N(0,1).$$

$$P(120 < Y < 130) = P\left(\frac{120-100}{\sqrt{90}} < \frac{Y-100}{\sqrt{90}} < \frac{130-100}{\sqrt{90}}\right)$$

Convert to CDF values:

$$P(120 < Y < 130) \approx \Phi\left(\frac{130-100}{\sqrt{90}}\right) - \Phi\left(\frac{120-100}{\sqrt{90}}\right)$$

$$= \Phi(3.1623) - \Phi(2.1082) \approx 0.01673.$$

(Check: exact binomial sum

$$\sum_{k=121}^{129} \binom{1000}{k} 0.1^k 0.9^{1000-k} \approx 0.01592,$$

which lies between the two CLT estimates.)

Normal approx

Too high

Central Limit Theorem (CLT)

Continuity correction

When we use the **Central Limit Theorem**, we replace the **discrete** binomial random variable Y (which can take only integer values) with a **continuous** normal random variable.

However, the probability

$$P(a < Y < b)$$

for a discrete Y represents the sum of individual point masses:

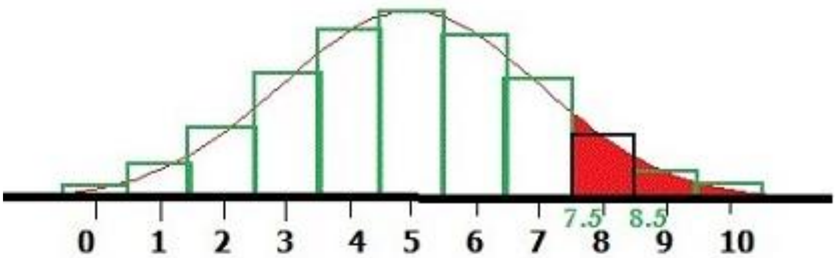
$$P(a < Y < b) = P(Y = a+1) + P(Y = a+2) + \cdots + P(Y = b-1).$$

In the continuous approximation, these point probabilities are represented by **areas under the normal curve**.
To make the continuous normal area better match the discrete sum, we "extend" the interval by **0.5 units** on each side:

- Replace $Y \geq a+1$ with $Y > a + 0.5$,
- Replace $Y \leq b-1$ with $Y < b - 0.5$.

Hence,

$$P(a < Y < b) = P(a+1 \leq Y \leq b-1) \;\approx\; P(a+0.5 < Y < b-0.5)$$



This small shift — called the **continuity correction** — significantly improves the accuracy of the normal approximation when dealing with discrete random variables such as the binomial.

Central Limit Theorem (CLT)

Example (with continuity correction CC)

In a communication system each data packet consists of 1000 bits. Due to the noise, each bit may be received in error with probability 0.1. It is assumed bit errors occur independently. Find the probability that there are more than 120 errors in a certain data packet and less than 130 errors.

$$P(120 < Y < 130) = P(121 \le Y \le 129) \approx \Phi\left(\frac{129.5-100}{\sqrt{90}}\right) - \Phi\left(\frac{120.5-100}{\sqrt{90}}\right)$$

$$= \Phi(3.1096) - \Phi(2.1609) \approx \boxed{0.01442.}$$

(Check: exact binomial sum

$$\sum_{k=121}^{129} \binom{1000}{k} 0.1^k 0.9^{1000-k} \approx 0.01592,$$

which lies between the two CLT estimates.)

Normal approx

Too low

Why CC didn't help much here because:

- $n = 1000$ is large,
- $p = 0.1$ is not too extreme,
- the interval $120 < Y < 130$ is *well inside* the distribution's main body (not near 0 or 1000).

In this case, the binomial distribution is already very smooth — so the discrete jumps are small.
That means the normal approximation (even *without* the correction) is already nearly perfect.

Central Limit Theorem (CLT)

Example

In a communication system each data packet consists of 20 bits. Due to the noise, each bit may be received in error with probability 0.1. It is assumed bit errors occur independently. Find the probability that there are more than 2 errors in a certain data packet and less than 4 errors.

Let Y be the number of bit errors in a 20-bit packet, with independent bit error probability $p = 0.1$.

Then $Y \sim \text{Binomial}(n = 20, p = 0.1)$.

$$E[Y] = np = 2, \qquad \text{Var}(Y) = np(1-p) = 1.8, \qquad \sqrt{\text{Var}(Y)} = \sqrt{1.8}$$

CLT (no continuity correction)

$$P(2 < Y < 4) = P\left(\frac{2-2}{\sqrt{1.8}} < \frac{Y-2}{\sqrt{1.8}} < \frac{4-2}{\sqrt{1.8}}\right) \approx \Phi\left(\tfrac{4-2}{\sqrt{1.8}}\right) - \Phi\left(\tfrac{2-2}{\sqrt{1.8}}\right)$$

$$P(2 < Y < 4) \approx \Phi(1.490712) - \Phi(0) = 0.431981.$$

CLT with continuity correction

$$P(Y = 3) \approx P(2.5 < Y < 3.5) = P\left(\frac{2.5-2}{\sqrt{1.8}} < \frac{Y-2}{\sqrt{1.8}} < \frac{3.5-2}{\sqrt{1.8}}\right)$$

$$P(Y = 3) \approx \Phi(1.118034) - \Phi(0.372678) = 0.222918.$$

Normal approx

Much closer

Exact binomial probability

$$P(Y = 3) = \binom{20}{3}(0.1)^3(0.9)^{17} = 1140 \cdot 0.001 \cdot 0.9^{17} = \boxed{0.1901198714}.$$

Estimator and Estimate

We want to learn about an **unknown parameter** θ. Example: the true proportion of voters supporting Candidate A.

- θ is fixed, but **unknown**.
- We collect data: $X_1, X_2, \ldots, X_n$.
- From the data, we compute an **estimator** $\hat{\Theta}$.

Example: In a poll with $n = 100$ voters:

- True (but unknown) proportion = θ.
- Estimator = $\hat{\Theta} = Y/n$, where Y is number of people in sample supporting Candidate A.
- After observing data, we compute $\hat{\theta}$, e.g. 53/100 = 0.53.

So:

- θ: fixed, unknown parameter.
- $\hat{\Theta}$: estimator (a random variable, depends on the random sample).
- After performing the experiment, we obtain a realized value denoted by $\hat{\theta}$ (the **estimate**).

i.i.d. and Sampling

To analyze data mathematically, we assume we have a **random sample**:

> **Random sample (i.i.d.)**: $X_1, \ldots, X_n$ are *independent* and *identically distributed* with the same distribution F_X.

This means:

- Each X_i has the same distribution.
- Each X_i is chosen independently.

In **practice**, we often draw **without replacement**, but if the population is large, the difference is negligible compared to drawing with replacement.

In **theory**, we imagine that we draw **with replacement**. This ensures that every draw is **independent** of the others, which makes the mathematics much simpler.

Sample Mean

Estimating the mean

Suppose we want the **average (mean)** of a population:

- True mean: $\mu = \mathbb{E}[X]$.
- Estimator: **sample mean** $\hat{\Theta} = \bar{X} = \frac{X_1+X_2+\cdots+X_n}{n}$.

Here $\hat{\Theta}$ is random because it depends on the random sample.

After performing the experiment, we obtain the **estimate** $\hat{\theta}$, a numerical value for the population mean.

Properties of the sample mean (assuming i.i.d.)

1. **Expectation:** $\mathbb{E}[\overline{X}] = \mu$.
2. **Variance:** $\mathrm{Var}(\overline{X}) = \sigma^2/n$.
3. **Law of Large Numbers:** as n grows, $\overline{X}$ gets closer to μ.
4. **Central Limit Theorem:** for large n, $\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$.

Interpretation:

- More data → estimate more stable.
- With enough data, $\bar{X}$ behaves like a normal distribution.

Quiz

Example: In a poll with $n = 100$ voters:

- True (but unknown) proportion = θ.
- Estimator = $\hat{\Theta} = Y/n$, where Y is number of people in sample supporting Candidate A.

How is X_i distributed in this example?

A. $X_i \sim \mathrm{Bernoulli}(\theta)$

B. $X_i \sim \mathrm{Binomial}(n, \theta)$

C. $X_i \sim N(\theta, \theta(1-\theta))$

D. $X_i \sim \mathrm{Bernoulli}(1/2)$

E. $X_i \sim \mathrm{Uniform}(0, 1)$

Bias

Evaluating estimators — how to tell if $\hat{\Theta}$ is good

Defining bias

For a point estimator $\hat{\Theta}$ of a parameter θ, the **bias** is defined as

$$B(\hat{\Theta}) = E[\hat{\Theta}] - \theta.$$

- If $B(\hat{\Theta}) = 0$ **for all possible values of** θ, then $\hat{\Theta}$ is called an **unbiased point estimator** of θ.

Example

sample mean is unbiased

If $X_1, X_2, \dots, X_n$ are i.i.d random variables with **mean μ**, then the expectation of the **sample mean** is

$$E[\bar{X}] = \mu \;\Rightarrow E[\bar{X}] - \mu = 0$$

Quiz

If $X_1, X_2, \dots, X_n$ are i.i.d random variables with mean μ, then the expectation of X_1 is

$$E[X_1] = \mu = E[\bar{X}]$$

*Since this is an **unbiased estimator like sample mean,** both estimates must be equally good!*

A: True because they both “hit the target on average”

B: True because they have equal variance

C: False because we cannot expect X_1 to be the best representative for all n random variables.

D: False because they do not have equal variance

Quiz

Suppose $\hat{\Theta}_1$ is an estimator for θ with

$$E[\hat{\Theta}_1] = a\theta + b, \quad a \neq 0.$$

Which of the following choices for $\hat{\Theta}_2$ makes it an **unbiased estimator** for θ?

A. $\hat{\Theta}_2 = a\hat{\Theta}_1 + b$

B. $\hat{\Theta}_2 = \dfrac{\hat{\Theta}_1}{a} + b$

C. $\hat{\Theta}_2 = \dfrac{\hat{\Theta}_1 - b}{a}$

D. $\hat{\Theta}_2 = \dfrac{\hat{\Theta}_1}{a - b}$

Bias

Example: Making a Biased Estimator Unbiased

Suppose $\hat{\Theta}_1$ is an estimator for θ with

$$E[\hat{\Theta}_1] = a\theta + b, \quad a \neq 0.$$

because the **expectation operator is linear**, any linear transformation of $\hat{\Theta}_1$ will lead to a predictable, linear change in its mean.
Therefore, if the bias in the mean is linear, a **linear correction** is sufficient to remove it.

We thus consider a general linear correction:

$$\hat{\Theta}_2 = \alpha\hat{\Theta}_1 + \beta,$$

where α and β are constants to be determined.

Taking expectations:

$$E[\hat{\Theta}_2] = \alpha E[\hat{\Theta}_1] + \beta = \alpha(a\theta + b) + \beta.$$

For $\hat{\Theta}_2$ to be unbiased, we require $E[\hat{\Theta}_2] = \theta$ for all θ.
Matching coefficients gives

$$\alpha a = 1, \qquad \alpha b + \beta = 0.$$

Hence

$$\alpha = \frac{1}{a}, \quad \beta = -\frac{b}{a}.$$

The unbiased estimator is therefore

$$\boxed{\hat{\Theta}_2 = \frac{\hat{\Theta}_1 - b}{a}}.$$

Bias and Variance

Mean Squared Error (MSE)

The mean squared error (MSE) of a point estimator $\hat{\Theta}$ is defined as

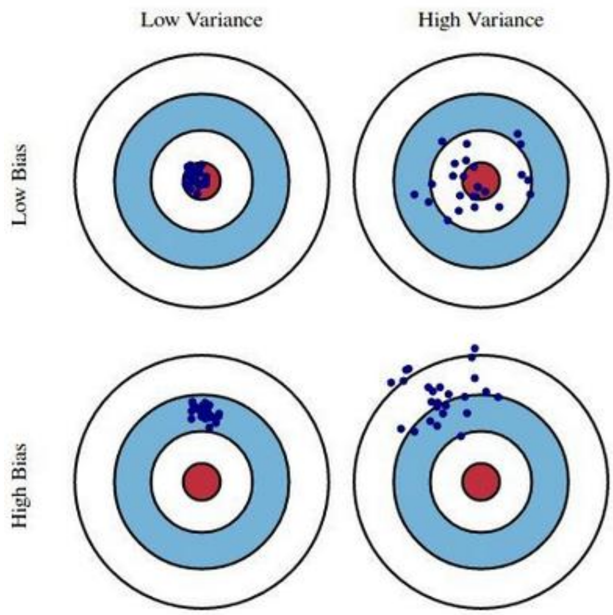
$$MSE(\hat{\Theta}) = E[(\hat{\Theta} - \theta)^2].$$

Decomposition into Variance and Bias

$$\begin{aligned} MSE(\hat{\Theta}) &= E\big[(\hat{\Theta} - \theta)^2\big] \\ &= \mathrm{Var}(\hat{\Theta} - \theta) + \big(E[\hat{\Theta} - \theta]\big)^2 \\ &= \mathrm{Var}(\hat{\Theta}) + \big(E[\hat{\Theta}] - \theta\big)^2 \\ &= \mathrm{Var}(\hat{\Theta}) + B(\hat{\Theta})^2, \end{aligned}$$

So, the MSE combines both:

- **Variance** → how much the estimator fluctuates.
- **Bias**^{2} → how far the estimator is shifted away from the true value.



Evaluating Point Estimators

Example: Mean Squared Error

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean

$$E[X_i] = \theta$$

and variance

$$\mathrm{Var}(X_i) = \sigma^2.$$

Consider two estimators for θ:

1. $\hat{\Theta}_1 = X_1$,
2. $\hat{\Theta}_2 = \bar{X} = \dfrac{X_1 + X_2 + \cdots + X_n}{n}$.

Estimator 1: $\hat{\Theta}_1 = X_1$

$$E[\hat{\Theta}_1] = E[X_1] = \theta \quad \Rightarrow \quad B(\hat{\Theta}_1) = 0.$$

$$\mathrm{Var}(\hat{\Theta}_1) = \mathrm{Var}(X_1) = \sigma^2.$$

$$\Rightarrow MSE(\hat{\Theta}_1) = \mathrm{Var}(\hat{\Theta}_1) + B(\hat{\Theta}_1)^2 = \sigma^2 + 0 = \sigma^2.$$

Estimator 2: $\hat{\Theta}_2 = \bar{X}$

$$E[\hat{\Theta}_2] = E[\bar{X}] = \theta \quad \Rightarrow \quad B(\hat{\Theta}_2) = 0.$$

$$\mathrm{Var}(\hat{\Theta}_2) = \mathrm{Var}\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{\sigma^2}{n}.$$

$$\Rightarrow MSE(\hat{\Theta}_2) = \mathrm{Var}(\hat{\Theta}_2) + B(\hat{\Theta}_2)^2 = \frac{\sigma^2}{n} + 0 = \frac{\sigma^2}{n}.$$

Comparison

For $n > 1$:

$$MSE(\hat{\Theta}_1) = \sigma^2 \quad > \quad \frac{\sigma^2}{n} = MSE(\hat{\Theta}_2).$$

Evaluating Point Estimators

Estimator for variance when μ is known

Let $X_1, \dots, X_n$ be i.i.d. with $E[X_i] = \mu$ (known) and $\mathrm{Var}(X_i) = \sigma^2$

Set $g(x) = (x-\mu)^2$.

Then

$$\sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2] = E[g(X)]$$

Since σ^2 is the expectation of $g(X)$, estimate it by the **sample mean** of $g(X_k)$:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{k=1}^{n}(X_k-\mu)^2$$

Check for **Unbiasedness**

$$\begin{aligned}
E[\hat{\sigma}^2] &= \frac{1}{n}\sum_{k=1}^{n} E\left[(X_k-\mu)^2\right] \\
&= \frac{1}{n}\sum_{k=1}^{n} E\left[X_k^2 - 2\mu X_k + \mu^2\right] \\
&= \frac{1}{n}\sum_{k=1}^{n} \left(E[X_k^2] - 2\mu\, E[X_k] + \mu^2\right) \\
&= \frac{1}{n}\sum_{k=1}^{n} \left(E[X_k^2] - 2\mu^2 + \mu^2\right) \qquad (\text{since } E[X_k] = \mu) \\
&= \frac{1}{n}\sum_{k=1}^{n} \left(E[X_k^2] - \mu^2\right) \\
&= \frac{1}{n}\sum_{k=1}^{n} \mathrm{Var}(X_k) \qquad \left(\mathrm{Var}(X_k) = E[X_k^2] - \mu^2\right) \\
&= \frac{1}{n}\sum_{k=1}^{n} \sigma^2 = \sigma^2.
\end{aligned}$$

Evaluating Point Estimators

Variance when μ is unknown

In practice, we rarely know μ. Instead, we estimate it with $\overline{X}$.

A natural idea is to replace μ with $\overline{X}$:

$$\overline{S}^2 = \frac{1}{n}\sum_{k=1}^{n}(X_k - \overline{X})^2.$$

Turns out to be biased:

$$E[\overline{S}^2] = \frac{n-1}{n}\sigma^2.$$

With bias given by:

$$B(\bar{S}^2) = E[\bar{S}^2] - \sigma^2 = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}.$$

Evaluating Point Estimators

Fixing the bias: the sample variance

To fix this, we slightly change the denominator from n to $n-1$:

$$S^2 = \frac{1}{n-1}\sum_{k=1}^{n}(X_k - \overline{X})^2.$$

This corrected version S^2 is **unbiased**:

$$E[S^2] = \sigma^2.$$

Proof that $E[S^2] = \sigma^2$

Expand first, then take expectations:

$$\sum_{k=1}^{n}(X_k - \bar{X})^2 = \sum_{k=1}^{n} X_k^2 - 2\bar{X}\sum_{k=1}^{n} X_k + n\bar{X}^2 = \sum_{k=1}^{n} X_k^2 - n\bar{X}^2,$$

since $\sum_{k=1}^{n} X_k = n\bar{X}$.

Now take expectations:

$$\mathbb{E}\left[\sum_{k=1}^{n}(X_k - \bar{X})^2\right] = \sum_{k=1}^{n}\mathbb{E}[X_k^2] - n\,\mathbb{E}[\bar{X}^2].$$

Because X_i are i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$,

$$\mathbb{E}[X_k^2] = \sigma^2 + \mu^2, \qquad \mathbb{E}[\bar{X}^2] = \mathrm{Var}(\bar{X}) + (\mathbb{E}[\bar{X}])^2 = \frac{\sigma^2}{n} + \mu^2.$$

Hence

$$\mathbb{E}\left[\sum_{k=1}^{n}(X_k - \bar{X})^2\right] = n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = (n-1)\sigma^2.$$

Dividing by $n-1$ gives $\mathbb{E}[S^2] = \sigma^2$.

Maximum Likelihood Estimation

Suppose we observe a sample from a distribution with parameter θ

$$X_1 = x_1,\ X_2 = x_2,\ \ldots,\ X_n = x_n.$$

Then the likelihood function is defined in general as

$$L(x_1, x_2, \ldots, x_n; \theta) = f_{X_1,\ldots,X_n}(x_1, x_2, \ldots, x_n; \theta).$$

If $X_1, X_2, \ldots, X_n$ are i.i.d., then $L(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} f_{X_i}(x_i; \theta)$.

The likelihood function is:

- **joint pmf/pdf of the observed data, viewed as a function of the parameter $\boldsymbol{\theta}$.**

The likelihood function evaluates:

- **how well different parameter values explain the observed data.**

Many books write likelihood as

$$L(\theta \mid x_1, \ldots, x_n)$$

to emphasize that the data are fixed.

But θ **is not random**, so we **cannot assign probabilities to it**.

So likelihood is **not a probability distribution over** θ.

Maximum Likelihood Estimation

The **maximum likelihood estimator** is given by

$$\hat{\Theta}_{ML} = \arg\max_{\theta} L(x_1, x_2, \ldots, x_n; \theta)$$

- $\max_\theta L(x_1, \ldots, x_n; \theta)$ means "the maximum value of the likelihood function." That's just a number.
- $\arg\max_\theta L(x_1, \ldots, x_n; \theta)$ means "the value(s) of θ where the maximum occurs." That's the **location of the maximum**, not the maximum itself.

Because logarithms are easier to work with, we often maximize the **log-likelihood**:

$$\ln L(x_1, x_2, \ldots, x_n; \theta).$$

Maximum Likelihood Estimation

Maximum Likelihood Estimation for the Normal Parameters

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with distribution

$$X_i \sim N(\theta_1, \theta_2), \qquad \theta_1 \in \mathbb{R},\ \theta_2 > 0.$$

That is, each X_i has density

$$f_{X_i}(x_i; \theta_1, \theta_2) = \frac{1}{\sqrt{2\pi\theta_2}} \exp\left(- \frac{(x_i - \theta_1)^2}{2\theta_2} \right).$$

Suppose we observe a sample $(x_1, x_2, \ldots, x_n)$.

Likelihood function

$$L(x_1, \ldots, x_n; \theta_1, \theta_2) = \frac{1}{(2\pi\theta_2)^{n/2}} \exp\left(- \frac{1}{2\theta_2} \sum_{i=1}^{n} (x_i - \theta_1)^2 \right).$$

Log-likelihood function

$$\ell(\theta_1, \theta_2) = \ln L = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\theta_2) - \frac{1}{2\theta_2} \sum_{i=1}^{n} (x_i - \theta_1)^2.$$

Differentiate and set to zero

- With respect to θ_1:

$$\frac{\partial \ell}{\partial \theta_1} = \frac{1}{\theta_2} \sum_{i=1}^{n} (x_i - \theta_1) = 0 \implies \hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

- With respect to θ_2:

$$\frac{\partial \ell}{\partial \theta_2} = -\frac{n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum_{i=1}^{n} (x_i - \theta_1)^2 = 0 \implies \hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\theta}_1)^2.$$

Maximum Likelihood Estimation

Example: MLE for a Normal Sample

Suppose we observe the data

$$x = (4.8,\ 5.2,\ 4.9,\ 5.1,\ 5.0).$$

Assume

$$X_i \sim N(\theta_1, \theta_2)$$

MLE estimators

$$\hat{\theta}_1 = \bar{x}, \qquad \hat{\theta}_2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Estimate the mean

$$\bar{x} = \frac{4.8 + 5.2 + 4.9 + 5.1 + 5.0}{5} = 5.0$$

$$\hat{\theta}_1 = 5.0$$

Estimate the variance

$$(x_i - \bar{x})^2 : \quad 0.04,\ 0.04,\ 0.01,\ 0.01,\ 0$$

$$\sum (x_i - \bar{x})^2 = 0.10$$

$$\hat{\theta}_2 = \frac{0.10}{5} = 0.02$$

Estimated model

$$X \sim N(5.0,\ 0.02)$$

The fitted normal distribution has **mean ≈ 5** and **very small variance**, meaning the observations are tightly clustered around 5.